

NIST AI RMF Profile: Amagi Framework Target State

Official Risk Management Framework Profile

Document Classification: Official NIST AI RMF Profile

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Profile Name: Amagi Framework Target Profile

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Legal Entity (Implementation): AGIAM Technologies Ltd

Jurisdiction: England and Wales

Role: Global Licensing and Certification Hub for the Amagi Framework

Commercial Licensing: Strict Architectural Principle License (S-APL), governed by the laws of England and Wales

1. Executive Summary

Profile Overview

This NIST AI RMF Profile documents the target state for [Amagi Framework v1.4](#), a revolutionary hardware-enforced AI safety architecture that provides mathematically guaranteed non-bypassability for high-risk AI systems. The framework implements a comprehensive risk management approach aligned with NIST AI RMF 1.0 principles, focusing on enterprise-grade AI safety solutions.

Strategic Objectives

The Amagi Framework target state aims to establish industry-leading AI risk management practices through:

- **Hardware-enforced safety guarantees** that eliminate software vulnerabilities
- **Mathematically proven non-bypassability** through TTP Triad architecture
- **Enterprise deployment readiness** for high-stakes AI applications
- **Regulatory compliance** with global AI safety standards

Key Outcomes

- **Risk Identification:** Real-time threat detection and classification
- **Risk Measurement:** Deterministic sub-millisecond response guarantees
- **Risk Management:** Hardware-level intervention capabilities
- **Governance:** Centralized security council with formal oversight

Business Impact

Target implementation has the potential to enable:

- Enterprise licensing opportunities in the indicative range of \$100K-1M+ per deployment
- Government contract eligibility for projects potentially exceeding \$50M
- Insurance premium reductions of 30-50% for certified AI systems
- Streamlined international certification pathway optimization

Competitive Advantage and Market Positioning

Amagi Framework occupies a unique position in the AI safety market with unprecedented regulatory compliance coverage:

EU AI Act Compliance Leadership:

- **Comprehensive Coverage:** Designed to achieve complete compliance with EU AI Act requirements including technical security (40%) AND organizational governance (60%)
- **Market Gap Analysis:** Existing competitors (AIM Security, Monitaur, Holistic AI, Lasso Security) achieve only 30-40% compliance coverage, focusing solely on technical security aspects
- **Structural Advantage:** Amagi's hardware-enforced architecture provides organizational governance guarantees impossible with software-only solutions

Competitive Differentiation:

- **Comprehensive vs Fragmented:** While competitors offer point security solutions, Amagi provides integrated compliance framework
- **Hardware vs Software:** Amagi's structural guarantees through TTP Triad cannot be bypassed, unlike software security layers
- **Regulatory vs Technical:** Full coverage of legal, organizational, and governance requirements beyond pure technical measures

Market Positioning:

- **Unique Category:** First and only comprehensive AI compliance framework providing structural guarantees
 - **Regulatory Moat:** Superior EU AI Act compliance creates sustainable competitive advantage
 - **Enterprise Ready:** Ready for immediate deployment in high-stakes environments requiring full regulatory compliance
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2. Background

Purpose and Scope

This profile defines the target state for Amagi Framework AI risk management practices, establishing a comprehensive framework for deploying hardware-enforced AI safety solutions across enterprise environments. The profile covers all aspects of AI system lifecycle management from initial design through operational deployment.

Framework Architecture

[Amagi Framework v1.4](#) implements a novel hardware-enforced architecture consisting of:

Core Components:

- **TTP Triad:** Trigger-Programmable Delay-Localized Activation mechanism providing hardware-level security guarantees
- **NIPU:** Neuro-Immune Processing Unit [DOI: 10.5281/zenodo.17682668](#) enabling isolated cognitive computation
- **MEM:** Memory Fabric [DOI: 10.5281/zenodo.17679327](#) with self-healing capabilities and continuous monitoring
- **SCC:** Security Council Core as centralized governance authority
- **MAG:** Memory Access Graph defining immutable policy matrix
- **TDF:** Trigger-Delay Fabric enforcing mandatory temporal separation
- **Amagi-Q Engine:** Quantum-resistant cryptographic processing unit

Target Profile Justification

A "Target" profile is appropriate for Amagi Framework because:

- Current state represents proof-of-concept implementation
 - Target state represents full enterprise deployment capability
 - Allows systematic progression toward industry-leading practices
 - Enables strategic planning for international certification
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3. Intended Audience

Primary Stakeholders

1. Executive Leadership

- Chief Technology Officers
- Chief Risk Officers
- Chief Security Officers
- Board Members and Investors

2. Technical Leadership

- AI Architects and Engineers
- Security Engineers
- Compliance Officers
- System Integration Specialists

3. Regulatory and Standards Bodies

- NIST AI Program Office
- ISO Technical Committee 164
- EU AI Act Conformity Assessment Bodies
- Industry-specific regulatory agencies

4. Enterprise Customers

- Healthcare organizations requiring AI compliance
- Automotive manufacturers (ISO 26262)
- Aerospace and defense contractors
- Financial services institutions

5. Certification Bodies

- Third-party assessors
- Insurance companies
- Audit firms
- Standards organizations

Usage Guidelines

This profile serves as:

- **Implementation roadmap** for technical teams
 - **Compliance guide** for regulatory assessment
 - **Business case** for executive decision-making
 - **Reference standard** for partner organizations
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4. AI System Management and Schedule

Organizational Structure

Security Council Core (SCC) - Central Authority

- **Decision Authority:** Final approval for all AI system modifications
- **Composition:** Chief Risk Officer, Chief Security Officer, AI Safety Director
- **Meeting Frequency:** Weekly operational reviews, monthly strategic assessments
- **Decision Process:** Consensus-based with technical advisory input

Technical Implementation Teams

- **Architecture Team:** TTP Triad and NIPU development
- **Integration Team:** MEM and MAG system integration

- **Security Team:** TDF and SCC security implementation
- **Compliance Team:** Regulatory alignment and certification

AI System Lifecycle Stages

Stage 1: Design and Architecture (Months 1-3)

- Requirements analysis and threat modeling
- TTP Triad architecture validation
- NIPU integration planning
- Security council initial review

Stage 2: Development and Integration (Months 4-8)

- Core component implementation
- MEM fabric deployment
- SCC integration and testing
- Security testing and validation

Stage 3: Verification and Validation (Months 9-12)

- Formal verification of TTP Triad
- Performance benchmarking
- Security penetration testing
- Regulatory compliance verification

Stage 4: Deployment and Operations (Months 13+)

- Production deployment
- Continuous monitoring implementation
- Incident response procedures
- Regular security assessments

Stakeholder Engagement Schedule

Monthly Stakeholder Reviews

- Progress assessment against target profile
- Risk assessment updates
- Compliance milestone tracking
- Resource allocation adjustments

Quarterly Strategic Reviews

- Target profile alignment assessment
- Industry standard updates integration
- Regulatory requirement changes
- Business strategy adjustments

Annual Comprehensive Assessment

- Full NIST AI RMF compliance audit
 - Target profile updates and revisions
 - Strategic planning for next year
 - Certification renewal planning
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5. Prioritization of Subcategories

Priority Level 1: Critical (Immediate Implementation)

GOVERN Function Priority:

- **GOVERN 1.1:** AI governance policies and procedures - TTP Triad as policy enforcement
- **GOVERN 2.1:** Roles and responsibilities - SCC clear authority definition
- **GOVERN 3.1:** Diverse team representation - MAG policy matrix governance

MAP Function Priority:

- **MAP 1.1:** AI system context definition - Hardware isolation boundaries
- **MAP 1.2:** AI actor identification - Real-time actor monitoring
- **MAP 1.3:** AI system risk categorization - 5 cognitive cluster classification

MEASURE Function Priority:

- **MEASURE 1.1:** AI system risk measurement - deterministic sub-millisecond response timing
- **MEASURE 1.2:** AI system performance monitoring - MEM continuous tracking
- **MEASURE 1.3:** AI system bias characterization - Hardware bias prevention

MANAGE Function Priority:

- **MANAGE 1.1:** AI system risk response - TTP Triad immediate intervention
- **MANAGE 1.2:** AI system risk prioritization - Priority-based delay programming
- **MANAGE 1.3:** AI system risk communication - SCC central communication

Priority Level 2: Important (6-month Implementation)

GOVERN Function:

- **GOVERN 1.2:** AI risk management strategy documentation
- **GOVERN 1.3:** AI risk tolerance and criteria establishment
- **GOVERN 4.2:** Risk and impact documentation processes
- **GOVERN 5.1:** External stakeholder feedback integration
- **GOVERN 6.1:** Third-party risk management policies

MAP Function:

- **MAP 1.4:** AI system harm identification and categorization
- **MAP 2.1:** AI system data governance mapping
- **MAP 2.2:** AI system process mapping

MEASURE Function:

- **MEASURE 1.4:** AI system risk management effectiveness evaluation
- **MEASURE 2.1:** AI system testing and validation metrics
- **MEASURE 2.2:** AI system monitoring and review metrics

MANAGE Function:

- **MANAGE 1.4:** AI system risk monitoring and review procedures
- **MANAGE 2.1:** AI system change management protocols
- **MANAGE 2.2:** AI system incident response procedures

Priority Level 3: Enhancement (12-month Implementation)

GOVERN Function:

- **GOVERN 1.4:** AI portfolio management procedures
- **GOVERN 1.5:** AI system lifecycle management policies
- **GOVERN 1.6:** AI system sunset procedures
- **GOVERN 1.7:** AI system oversight and accountability measures
- **GOVERN 2.2:** AI personnel training and awareness programs

MAP Function:

- **MAP 2.3:** AI system impact assessment mapping
- **MAP 2.4:** AI system interdependency mapping

MEASURE Function:

- **MEASURE 2.3:** AI system audit and assessment metrics
- **MEASURE 2.4:** AI system external evaluation metrics

MANAGE Function:

- **MANAGE 2.3:** AI system improvement and enhancement procedures
 - **MANAGE 2.4:** AI system retirement and decommissioning procedures
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6. Use Cases

Use Case 1: Medical AI Systems

Context: High-risk medical AI applications requiring regulatory compliance

System Characteristics:

- Patient data processing and analysis
- Diagnostic decision support
- Treatment recommendation systems
- Real-time monitoring and alerts

Risk Profile:

- **High Impact:** Patient safety implications
- **Regulatory Requirements:** FDA, EU MDR, HIPAA compliance
- **Technical Requirements:** 99.99% uptime, <10ms response time
- **Security Requirements:** HIPAA-compliant data handling

Amagi Framework Implementation:

- **TTP Triad:** Hardware-enforced patient safety protocols
- **NIPU:** Continuous medical device verification
- **MEM:** Secure patient data isolation and processing
- **SCC:** Medical device governance and oversight
- **TDF:** Real-time safety enforcement for patient protection

Target Outcomes:

- Zero tolerance for safety violations
- Real-time threat detection and response

- Regulatory compliance documentation
- Patient safety guarantee mechanisms

Use Case 2: Autonomous Vehicle AI

Context: Safety-critical autonomous driving systems

System Characteristics:

- Real-time perception and decision-making
- Vehicle control and navigation
- V2X communication protocols
- Passenger and pedestrian safety

Risk Profile:

- **High Impact:** Human life safety implications
- **Regulatory Requirements:** ISO 26262, NHTSA guidelines
- **Technical Requirements:** Sub-millisecond response time, 99.999% reliability
- **Security Requirements:** Vehicle cybersecurity standards

Amagi Framework Implementation:

- **TTP Triad:** Hardware-enforced driving safety protocols
- **NIPU:** Continuous vehicle system verification
- **MEM:** Secure sensor data processing and fusion
- **SCC:** Vehicle safety governance and decision authority
- **TDF:** Real-time safety enforcement for driving protection

Target Outcomes:

- Guaranteed safety intervention capabilities
- Real-time threat detection and mitigation
- Regulatory compliance with automotive standards
- Passenger safety guarantee mechanisms

Use Case 3: Financial Services AI

Context: High-stakes financial decision-making systems

System Characteristics:

- Real-time trading and risk assessment
- Fraud detection and prevention
- Regulatory compliance monitoring
- Customer data protection

Risk Profile:

- **High Impact:** Financial loss and regulatory penalties
- **Regulatory Requirements:** Basel III, PCI DSS, GDPR
- **Technical Requirements:** Sub-millisecond latency, 99.9% availability
- **Security Requirements:** Bank-grade security and encryption

Amagi Framework Implementation:

- **TTP Triad:** Hardware-enforced financial transaction safety

- **NIPU:** Continuous financial system verification
- **MEM:** Secure financial data isolation and processing
- **SCC:** Financial governance and compliance oversight
- **TDF:** Real-time safety enforcement for financial protection

Target Outcomes:

- Zero tolerance for unauthorized transactions
- Real-time threat detection and prevention
- Regulatory compliance documentation
- Financial data protection guarantee mechanisms

Use Case 4: Aerospace and Defense AI

Context: Mission-critical aerospace and defense applications

System Characteristics:

- Autonomous flight control systems
- Military decision support systems
- Space mission planning and execution
- Critical infrastructure protection

Risk Profile:

- **Extreme Impact:** National security and human life
- **Regulatory Requirements:** FAA, DoD guidelines, ITAR
- **Technical Requirements:** Military-grade reliability and security
- **Security Requirements:** Highest level of cybersecurity

Amagi Framework Implementation:

- **TTP Triad:** Hardware-enforced mission safety protocols
- **NIPU:** Continuous aerospace system verification
- **MEM:** Secure mission data isolation and processing
- **SCC:** Aerospace governance and mission oversight
- **TDF:** Real-time safety enforcement for mission protection

Target Outcomes:

- Guaranteed mission safety intervention capabilities
 - Real-time threat detection and mitigation
 - Regulatory compliance with aerospace standards
 - Mission success guarantee mechanisms
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7. Pathway to Target State Implementation

Implementation Framework

The Target State outlined in this profile is designed for incremental adoption across enterprise environments. The [Amagi Framework v1.4](#)'s modular architecture, centered on

the TTP Triad Hard Macro IP, enables organizations to integrate structural safety guarantees into existing systems with minimal disruption.

Deployment Timeline

Phase 1 (1-2 weeks): TTP Triad integration with existing AI systems

- Hardware-level safety mechanism deployment
- Integration with current security infrastructure
- Basic threat detection and response capability

Phase 2 (3-6 months): SCC governance implementation and MEM deployment

- Security Council Core operational procedures
- Memory fabric isolation and self-healing capabilities
- Comprehensive governance framework activation

Phase 3 (6-12 months): Full target profile realization for high-stakes applications

- Complete TTP Triad architectural implementation
- Full NIST AI RMF alignment achievement
- Regulatory compliance certification preparation

Phase 4 (12-18 months): International certification and enterprise-wide deployment

- ISO/IEC 42001 certification completion
- Global market deployment and standardization
- Continuous improvement and evolution

Adoption Readiness

The framework is designed for immediate integration with standard enterprise development cycles, requiring no fundamental architectural changes to existing AI infrastructure while providing hardware-level safety guarantees impossible to achieve through software-only solutions.

Appendix A: Technical Architecture Details

TTP Triad (Trigger-Programmable Delay-Localized Activation)

The TTP Triad forms the core security mechanism of [Amagi Framework v1.4](#), providing hardware-enforced guarantees for AI system security:

Trigger Component:

- Initiates security validation for all AI system actions
- Implements hardware-enforced triggers for security events
- Provides deterministic activation of safety mechanisms
- Creates immediate response capabilities for threat detection

Programmable Delay Component:

- Implements configurable safety delays for AI operations

- Provides flexible timing controls for different risk levels
- Enables priority-based response scheduling
- Creates configurable security checkpoints

Localized Activation Component:

- Ensures isolated activation of security mechanisms
- Provides hardware-enforced spatial separation
- Prevents system-wide disruptions from localized issues
- Creates targeted intervention capabilities

NIPU (Neuro-Immune Processing Unit) [DOI: 10.5281/zenodo.17682668](https://doi.org/10.5281/zenodo.17682668)

NIPU provides seamless security verification without impacting system performance:

Real-time Verification:

- Continuous verification of AI system integrity
- Non-intrusive security monitoring and validation
- Automated detection of security policy violations
- Immediate alerting for security anomalies

Mathematical Guarantees:

- Formal proofs of system security properties
- Verified implementation of security protocols
- Cryptographically secure verification mechanisms
- Tamper-evident security monitoring

MEM (Memory Fabric) [DOI: 10.5281/zenodo.17679327](https://doi.org/10.5281/zenodo.17679327)

MEM provides secure, self-healing memory management for AI systems:

Security Isolation:

- Hardware-enforced memory isolation between AI components
- Secure memory allocation and deallocation
- Protected memory regions for sensitive data
- Memory access control and monitoring

Self-healing Capabilities:

- Automatic detection and correction of memory errors
- Dynamic memory reconfiguration for optimal performance
- Memory leak detection and remediation
- Continuous memory health monitoring

SCC (Security Council Core)

SCC provides centralized governance and decision-making authority:

Central Authority:

- Single point of decision authority for all AI system modifications

- Formal governance processes and procedures
- Executive oversight for all security decisions
- Accountability framework for AI system outcomes

Risk Management:

- Continuous risk assessment and monitoring
- Decision-making authority for risk mitigation
- Communication hub for security information
- Incident response coordination and management

MAG (Memory Access Graph)

MAG provides immutable policy matrix defining all permitted data flows:

Policy Enforcement:

- 128×128 OTP matrix defining fine-grained, typed access rules
- Hardware-level policy decision on every access attempt
- Immutability ensures no runtime policy modification

TDF (Trigger-Delay Fabric)

TDF provides hardware-enforced temporal separation:

Temporal Security:

- Fixed hardware-enforced latency (1-10 μs)
- Prevents race conditions from Untrusted Computing Base
- Ensures mandatory pause before critical actions

Amagi-Q Engine

Amagi-Q provides quantum-resistant cryptographic processing:

Quantum Resistance:

- Post-quantum cryptographic algorithms
- Quantum-resistant key exchange protocols
- Future-proof security against quantum computing threats
- Mathematical proofs of quantum resistance

High-Performance Cryptography:

- Hardware-accelerated cryptographic operations
 - Optimized performance for real-time security processing
 - Scalable cryptographic infrastructure
 - Integration with existing security frameworks
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Appendix B: Regulatory Compliance Mapping

NIST AI RMF 1.0 Alignment

This target profile demonstrates comprehensive alignment with NIST AI Risk Management Framework 1.0:

GOVERN Function Alignment:

- GOVERN 1.1-1.7: Complete policy and procedure framework
- GOVERN 2.1-2.2: Formal roles, responsibilities, and training
- GOVERN 3.1: Diverse team representation and decision-making
- GOVERN 4.2: Comprehensive risk and impact documentation
- GOVERN 5.1: External stakeholder feedback integration
- GOVERN 6.1: Third-party risk management policies

MAP Function Alignment:

- MAP 1.1-1.4: Complete AI system context and risk mapping
- MAP 2.1-2.4: Comprehensive data and process mapping
- MAP 3.1-3.2: Complete impact and interdependency mapping

MEASURE Function Alignment:

- MEASURE 1.1-1.4: Comprehensive risk measurement and monitoring
- MEASURE 2.1-2.4: Complete testing and evaluation metrics

MANAGE Function Alignment:

- MANAGE 1.1-1.4: Complete risk response and communication
- MANAGE 2.1-2.4: Comprehensive change and incident management

International Standards Compliance

ISO/IEC 42001 (AI Management Systems):

- Management system framework aligned with ISO requirements
- Risk management processes meeting ISO standards
- Continuous improvement mechanisms
- Management review and audit procedures

EU AI Act Compliance:

- High-risk AI system classification and management
- Conformity assessment preparation
- Risk management system implementation
- Transparency and human oversight requirements

ISO 26262 (Automotive Functional Safety):

- Safety integrity level (SIL) compliance
- Hazard analysis and risk assessment
- Safety validation and verification
- Functional safety management

Industry-Specific Standards:

- FDA Software as Medical Device (SaMD) guidance
 - NHTSA cybersecurity guidelines for vehicles
 - PCI DSS for financial services compliance
 - DoD cybersecurity framework for defense applications
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Appendix C: Implementation Timeline

Phase 1: Foundation (Months 1-3)

Month 1:

- Profile approval and stakeholder communication
- Resource allocation and team establishment
- Governance framework implementation
- Initial TTP Triad deployment

Month 2:

- NIPU verification system implementation
- MEM security fabric deployment
- SCC operational procedures establishment
- TDF threat detection system activation

Month 3:

- MAG policy matrix governance deployment
- Amagi-Q cryptographic engine integration
- Initial performance testing and validation
- Stakeholder review and feedback incorporation

Phase 2: Enhancement (Months 4-6)

Month 4:

- Priority Level 1 requirements completion
- Initial third-party assessment preparation
- Performance optimization and tuning
- Security penetration testing

Month 5:

- Priority Level 2 implementation begins
- Regulatory compliance documentation completion
- Industry engagement and partnership development
- Customer pilot program preparation

Month 6:

- Comprehensive system integration testing
- Pre-assessment evaluation and gap analysis

- International certification pathway planning
- Business development strategy finalization

Phase 3: Deployment (Months 7-12)

Months 7-9:

- Priority Level 2 completion
- Third-party certification assessment
- Customer pilot program execution
- Performance benchmarking and optimization

Months 10-12:

- Priority Level 3 implementation begins
- Full enterprise deployment preparation
- International certification pursuit
- Strategic planning for market launch

Success Criteria and Milestones

30-Day Milestones:

- Formal profile approval and implementation authorization
- Complete stakeholder alignment and communication
- Resource allocation and team establishment
- Initial governance framework operational

90-Day Milestones:

- Priority Level 1 requirements 100% complete
- Core system functionality validated and operational
- Initial performance benchmarks established
- Stakeholder confidence and support confirmed

6-Month Milestones:

- Priority Level 2 requirements 80% complete
- Third-party assessment readiness achieved
- Regulatory compliance documentation complete
- International certification pathway established

12-Month Milestones:

- Full target profile achievement (100% Priority Level 1 & 2)
- Third-party certification obtained
- Enterprise deployment capability demonstrated
- International market entry preparation complete

Document Control:

- **Classification:** Official NIST AI RMF Profile
- **Distribution:** Authorized stakeholders only
- **Review Cycle:** Quarterly updates with annual comprehensive review

- **Version Control:** Major updates require executive approval
- **Implementation Tracking:** Monthly progress reports and milestone tracking

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 - **Legal Entity:** AGIAM Technologies Ltd (to be incorporated in England and Wales)
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This NIST AI RMF Profile represents the target state for Amagi Framework AI risk management practices and serves as the primary reference for implementation, compliance, and certification activities.